

4. Setup, warm-up, and robot-to-sensor registration

Registration gives every later capture a ground-truth target pose in the sensor frame. Two things anchor it. Robot repeatability fixes relative motion. A plane-correspondence solve on the noise plate T2 fixes the camera's pose in the robot base, using the depth planes themselves. What the planes cannot observe, the sideways position of a target on the flange, comes from the as-built datum and a once-per-mount check against the left IR image (Step 4.8).

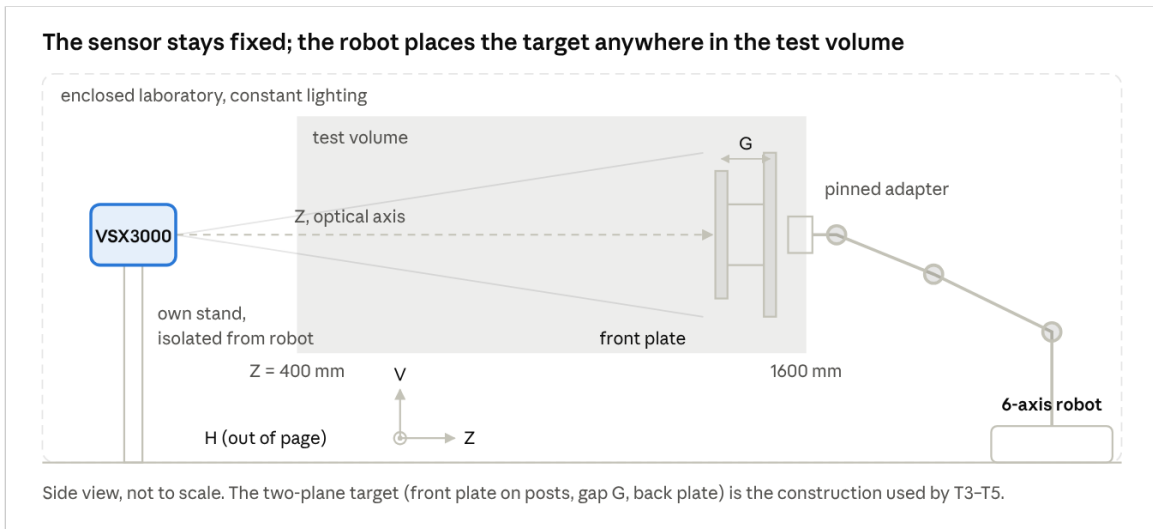


Figure 3. Experimental setup, side view

The arrangement is shown in Figure 3. After Step 4.7, the robot-base-to-sensor transform is known, so every commanded target pose is also a ground-truth pose in sensor coordinates.

1. Environment. The laboratory is enclosed and its lighting is constant; keep it unchanged for the whole session. Log the air and sensor-housing temperatures every `TEMPERATURE_LOG_INTERVAL_MIN`.
2. Sensor configuration. Disable auto-exposure and fix exposure, gain, emitter power, and trigger mode. Record every depth-processing setting (temporal filter, spatial filter, hole filling, confidence threshold) in `sensor_config.json`. Characterize the configuration that production will use. If the filters can be switched off, run A and B a second time with the filters off, so the sensor's own processing can be separated from its physics; this repeat is outside the Section 9 budget. Record the SDK and firmware versions in the same file.
3. Warm-up. Power the sensor for at least `WARMUP_MIN_MINUTES`. Then put T2 fronto-parallel at `Z_REFERENCE_MM` and capture `WARMUP_CHECK_FRAMES` frames every

WARMUP_CHECK_INTERVAL_MIN. Here σ_t is the temporal standard deviation of those warm-up frames themselves. Start testing once the mean plane Z has drifted less than $\text{WARMUP_DRIFT_FRACTION_OF_SIGMA} \times \sigma_t$ over $\text{WARMUP_DRIFT_WINDOW_MIN}$.

4. Settle and vibration check. With T2 at Z_MIN, where the depth noise is smallest and a robot vibration of a given amplitude is easiest to see, capture SETTLE_CHECK_FRAMES frames twice: once with the servos on after a move and the settle wait, and once with the brakes engaged. If the servo-on σ_t exceeds the brakes-on σ_t by more than $\text{SETTLE_SIGMA_EXCESS_FRACTION}$, raise ROBOT_SETTLE_TIME_S or stiffen the target mount, then repeat.

5. Intrinsic and frame checks.

1. Read the left IR intrinsics, the depth-to-IR extrinsics, SENSOR_BASELINE_MM, DEPTH_LSB_MM, the projector offset, and the frame rate from the SDK or datasheet. Fill in the † entries in Section 2. Confirm that the sensor returns valid depth on T2 at Z_MIN and at Z_MAX; if not, tighten Z_MIN_MM or Z_MAX_MM to the sensor's range and rebuild the station ladder.
 2. Confirm the depth image is registered to the left IR image. With T3a in view, overlay the square's edges from the IR image on the depth discontinuities. They must agree within FRAME_CHECK_PX.
 3. Confirm the baseline direction and side. The occlusion band (a strip of no-reads) appears beside vertical edges only if the baseline runs along H; if it appears beside horizontal edges, swap H and V in Procedures B and E. With the right camera at +H, as Section 1 assumes, the band lies outside the -H edge of the raised square and inside the +H edge of the window; if it lies on the other side, reverse the sign of H in Procedures B and E.
6. Registration capture. Mount T2. Command REGISTRATION_POSES poses that span Z_MIN to Z_MAX and cover the field of view, with tilts about H and about V, of both signs, up to REGISTRATION_TILT_RANGE_DEG, so that the plate normals span all three directions; the plate must stay inside the field at every tilt. At each pose capture REGISTRATION_FRAMES depth frames and the read-back robot pose.
7. Registration solve and acceptance. Fit the plate plane in each pose's mean depth frame (normal and distance in the camera frame). Solve the hand-eye problem from the plane correspondences: a closed-form start, then joint nonlinear least squares on the plane-normal and plane-distance residuals, giving the camera → robot-base transform and the plate's normal and offset on the flange. The plate's sideways position on the flange and its rotation about its normal are not observable from planes and are not needed. Accept if the RMS plane-distance residual is $\leq \text{REGISTRATION_RESIDUAL_ACCEPT_MM}$, and report beside it the standard error of the

camera's Z offset from the fit covariance, the conditioning that Section 15 estimates at about 0.1 mm. If either is too large, add poses with larger tilts, up to REGISTRATION_TILT_RANGE_DEG, and solve again. Save both transforms, the residual, and the standard error in registration.json.

8. Station targets and mount check. For every station in Sections 5–8, compute the robot pose that puts the target's reference point at the required (H, V, Z) in the camera frame, fronto-parallel unless stated otherwise. Re-mounting a target on the dowel-pinned adapter needs no new registration. Check each mount once at Z_REFERENCE_MM: fit the mounted target's front plane from the depth data and compare it with the registered pose, Z within REGISTRATION_RESIDUAL_ACCEPT_MM and tilt within MOUNT_TILT_TOLERANCE_DEG; then locate one feature edge or outline in the left IR image and compare its H and V with the as-built datum offsets, within FRAME_CHECK_PX. Otherwise re-seat the target or correct the datum record.